
EE/CprE/SE 491 WEEKLY REPORT 4

March 12 2025 - March 25 2025

Group number: 16

Project title: Outdoor Roomba

Client &/Advisor: Prof. Diane Rover

Team Members/Role:

Brodie M Bates

Daniel Vergara Pinilla

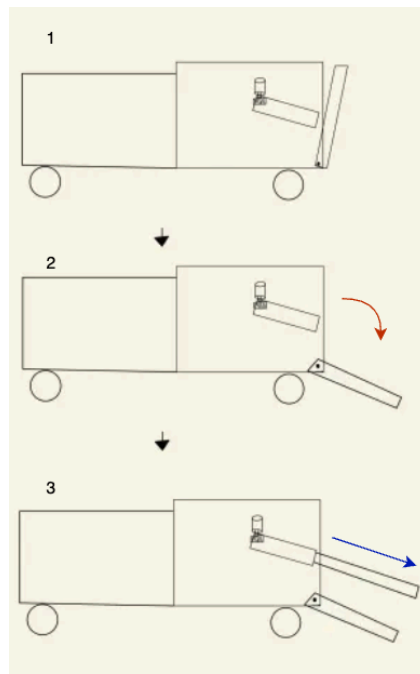
Mitchell D Owen

Om Shukla

Sudipta Halder

○ Weekly Summary

We needed a way to validate if our ideas for our scooping mechanism would be effective at collecting garbage. So we simulated our roomba on Webots, a platform open-source 3D robot simulation software. This way we were able to quickly test out our ideas. This week we just started to learn from tutorials on Webots and have been able to successfully create a basic four wheeler robot and was able to control it with a C script in simulation. During the meeting we brainstormed on the motion of the scooping mechanism. We decided to have two components: a lid that drops down to make a ramp for the trash to move into the compartment and arm like structure help move the trash to move up the ramp as rough illustrated below.



○ **Past week accomplishments**

- Sudipta Halder: Got started on following the tutorial on Webots. Made a basic robot on wheels and added the swapping mechanism to test out the swapping mechanism and effectiveness on it.
- Brodie Bates: Do Webots tutorial and create some more CAD parts for the robot. Worked with the team on bill of materials and designing the sweeper arm.
- Daniel Pinilla: Did research on multi-threading on raspberry and Websocket functionality
- Om Shukla: Learning Webots to help model robot functionality. Created a Trello board for keeping track of sprints.
- Mitchell Owen: This week was spent working on the 3D Modeling of the Outdoor Roomba. Deciding what we would do for the scooper and creating the sweeper all on Fusion.

○ **Pending issues**

- Brodie Bates: Need to get all of our ideas and designs for the robot on the same page as the rest of the team
- Mitchell Owen: Ensure everyone makes it to the in person meetings. For some reason, it is a struggle. Hopefully, the problem can be resolved otherwise, stuff isn't getting done.

○ **Individual contributions**

<u>NAME</u>	<u>Individual Contributions</u>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Brodie Bates	Redesigned extending out mechanism, started learning Webots, helping with bill of materials.	6	22
Om Shukla	Researched more on Webots and created a trello board	4	17
Sudipta Halder	Brainstormed some design decisions. Worked on simulation Webots. Checked out other robot simulation platforms on Unity, Isaac Sim Works.	10	25
Daniel Mauricio Vergara	Started AI code for self driving to trash	4	25
Mitchell Owen	Working on the 3D model on Fusion and filling in the Bill of Material	4	22

○ **Plans for the upcoming week**

- Sudipta Halder: Work on CAD modeling. Look into some RTOS that meet our requirements and support on raspberry PI. Get sensors and more components in the BOM. Populate a Trello board as a sprint board with current plans.
- Brodie Bates: Work on Webots with Sudipta, CAD model more parts, redesign extending out mechanism with the team, work on bill of materials and research where to get parts.
- Daniel Pinilla: Finish code for self driving
- Om Shukla: Learn more about Webots and model robot functionality like moving and object detection and object marking in a geofence
- Mitchell Owen: Getting a bill of materials finished so there is an idea of the price of everything for the outdoor Roomba. Also, learn more about some hardware that will be used.

○ **Summary of weekly advisor meeting**

In the meeting our advisor told us we should start putting together a bill of materials as well as start researching what software we are going to run on our raspberry pi. Software suggestion that she gave us was for FreeRTOS. We need to decide what we want to run if not FreeRTOS.